CSA1717-ARTIFICIAL INTELLIGENCE

CO2-AT3-DRY RUN TEST

PROBLEM STATEMENT

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Question 1:

Problem Statement

Consider the following weighted graph in which each edge represents the cost of moving from one node to another. The heuristic values h(n) indicate the estimated cost from each node to the goal node G.

```
          (A)
         /   \
     2 /      \ 4
      /         \
    (B)---3---(C)
     | \         |
   7 |  \2       | 3
     |    \       |
    (D)---2---(E)
      \
       \2
       (G)
```

Heuristic Values

Node	h(n)
A	7
B	6
C	4
D	3
E	2
G	0

Task

Perform a dry run of the A* Search Algorithm to find the shortest path from the Start Node (A) to the Goal Node (G).

For each iteration, display:

- Current Node
- Open List
- Closed List
- Actual Path Cost g(n)

- Heuristic Value h(n)
- Evaluation Function f(n) = g(n) + h(n)

Finally,

- Determine the optimal path from A to G.
- Calculate the total path cost.
- Explain how the heuristic values influence the search process and help in selecting the most promising node at each step.

Question 2:

Problem Statement

Consider the following two-player game tree. The root node represents the MAX player, while the second level represents the MIN player. The leaf nodes contain the utility values that determine the outcome of the game.

```
             MAX
            /      \
        MIN        MIN
      / |  \      / |  \
     3  5   6    9  1   2
```

Task

Perform a dry run of the Minimax Algorithm with Alpha-Beta Pruning by evaluating the game tree from left to right.

For each node, display:

- Alpha (α) value
- Beta (β) value
- Selected utility value
- Pruned node(s), if any

Finally,

- Determine the best move for the MAX player.
- Find the final minimax value.
- Identify the node(s) that are pruned using Alpha-Beta Pruning.
- Explain how Alpha-Beta Pruning reduces the number of node evaluations without affecting the optimal decision.